Math 301 - Project 2

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1 Abstract

The rotational dynamics of a spacecraft modeled as a single rigid body were analyzed to design a feedback controller capable of stabilizing its angular velocity about selected equilibrium orientations. The nonlinear equations of motion were linearized about three equilibrium points, and the resulting state-space representations were tested for controllability. Two equilibria, $\omega_{e1} = [1, 0, 0]^T$ and $\omega_{e2} = [0, 1, 0]^T$, were found to be fully controllable, while $\omega_{e3} = [0, 0, 1]^T$ was not able to controlled. I used the place() function to design a controller that forces the system's dynamics to have eigenvalues at those locations, making the system stable. Simulations verified that the controller successfully stabilized the spacecraft near the desired equilibrium, reduced angular velocity errors to zero, and required only modest torque inputs.

2 Introduction and Problem Description

The motion of a spacecraft can be described by a set of nonlinear rigid-body equations that relate angular velocity, control torques, and moments of inertia. Controlling these rotational motions is a fundamental problem in spacecraft design, as precise orientation is required for tasks such as antenna pointing, solar-array alignment, and scientific instrument stabilization. Even small disturbances or modeling errors can lead to undesired tumbling, making feedback control essential for maintaining a desired orientation.

In this project, the spacecraft was modeled as a symmetric rigid body with principal moments of inertia $J_1 = 0.0867$, $J_2 = 0.1908$, and $J_3 = 0.2708$. The three numbers $J_1 = 0.0867$, $J_2 = 0.1908$, and $J_3 = 0.2708$ come from the way the spacecraft's mass is distributed along its three axes. In this project, the spacecraft is modeled as a rectangular solid (a box-shaped rigid body) with side lengths of 1.5 m, 1.0 m, and 0.2 m and a total mass of 1 kg. Physically, the moment of inertia measures how difficult it is to spin an object about a given axis — the more mass spread out away from that axis, the larger the moment of inertia. Since the spacecraft is longest along one direction and shortest along another, each axis resists rotation to a different degree. Because the smallest dimension (0.2 m) contributes least to resistance, rotation about that axis is the easiest, giving the smallest inertia value $J_1 = 0.0867$. Rotation about an axis involving the longer dimensions requires more effort, producing larger values $J_2 = 0.1908$ and $J_3 = 0.2708$. The equations of motion were linearized about three candidate equilibrium angular velocities. Controllability analysis showed that rotations about the first two principal axes are fully controllable, while rotation about the third axis is not. A state-feedback controller using pole placement was then designed for the controllable equilibria, assigning closed-loop poles at -2.5, -3.0 and -3.5 to ensure stability with moderate response speed. Numerical simulations demonstrated that the controller successfully drives the angular velocity to the desired equilibrium from nearby initial conditions and maintains stability in the presence of external disturbances. These results illustrate the effectiveness of linear feedback control in stabilizing nonlinear spacecraft's dynamics.

3 Analysis With No Control

Before implementing feedback control, the natural stability properties of the spacecraft's rotational dynamics were analyzed under zero control input ($\tau_1 = \tau_2 = 0$). The nonlinear equations of motion exhibit equilibrium angular velocities when rotation occurs about one of the principal axes:

$$\omega_{e1} = [1, 0, 0]^T \quad \omega_{e2} = [0, 1, 0]^T \quad \omega_{e3} = [0, 0, 1]^T$$

Linearizing the system around each equilibrium can causes different coupling terms among the angular velocity components. From classical rigid-body dynamics, rotation about the largest and smallest moments of inertia (J_1 and J_3) is stable, while rotation about the intermediate axis (J_2) is unstable. Since the spacecraft has $J_1 < J_2 < J_3$, this predicts that the equilibria ω_{e1} and ω_{e3} should be stable, while ω_{e2} should be unstable.

These predictions were confirmed by numerical simulations using the Project2.m script without applied any related control. When the initial condition was chosen near ω_{e1} or ω_{e3}, the angular velocity components exhibited small oscillations but remained bounded, indicating neutral stability. In contrast, simulations initialized near ω_{e2} showed rapidly growing oscillations that caused the spacecraft to depart from the equilibrium orientation, consistent with theoretical instability about the intermediate axis. Overall, the behavior of our model without the control agrees with classical rigid-body stability theory and provides a clear motivation for designing a feedback controller to stabilize the otherwise unstable rotation about the intermediate axis.

4 Controllability Analysis

To determine whether the spacecraft's angular velocity can be stabilized about each equilibrium, we evaluted the linearized state-space model. Linearization of the nonlinear rigid-body equations about each equilibrium ω_e is:

$$\dot{x} = A(\omega_e)x + Bu \quad B = \begin{bmatrix} 1/J_1 & 0 \\ 0 & 1/J_2 \\ 0 & 0 \end{bmatrix}$$

where $x = \omega - \omega_e$ and the matrix A(ω_e) depends on the equilibrium angular velocity. The controllability matrix $\mathcal{C} = [B,\ AB,\ A^2B]$ was formed for each case, and its rank was computed using MATLAB. THis is the result that we get:

Equilibrium	**rank**	**Controllable or Not**
$\omega_{e1} = [1, 0, 0]^T$	3	Yes
$\omega_{e2} = [0, 1, 0]^T$	3	Yes
$\omega_{e3} = [0, 0, 1]^T$	2	No

The first two equilibria are fully controllable, meaning the spacecraft's angular velocity can be driven from any initial condition near these points to the desired equilibrium using appropriate torque inputs. The third equilibrium, corresponding to rotation about the w_3 axis, is not controllable because the control torques τ_1 and τ_2 do not directly or indirectly influence motion along that axis to first order. Physically, this occurs because there is no applied torque about the third principal axis, and the linearized coupling terms vanish for this orientation. Therefore, feedback control can be successfully applied only around ω_{e1} and ω_{e2}, while ω_{e3} remains uncontrollable in the linear approximation.

5 Control Design Scenario 1

For the first control design scenario, the equilibrium point $\omega_{e1} = [1, 0, 0]^T$. This is fully controllable and physically corresponds to a stable spin about the first principal axis. The linearized state-space model around this equilibrium has full rank (rank$(\mathcal{C}) = 3$), enabling the use of state-feedback control to regulate the angular velocity.

A feedback law of the form

$$u = -K(\omega - \omega_e)$$

was implemented, where $u = [\tau_1, \tau_2]^T$ and K was determined using MATLAB's place() function. The desired closed-loop poles were selected at -2.5, -3.0, and -3.5 to ensure a moderately fast but well-damped response. The resulting gain matrix was

$$K = \begin{bmatrix} 0.2600 & 0.0098 & -0.0766 \\ 0.0009 & 1.1450 & -4.1576 \end{bmatrix}$$

producing closed-loop eigenvalues that match the desired pole locations.

Simulations confirmed that this feedback successfully stabilizes the system around ω_{e1}. For small initial perturbations, the angular velocity smoothly converged to the desired equilibrium with both control torques decaying to zero. When the initial deviation was increased, convergence still occurred until the torque inputs reached their saturation limits ($|\tau_i| \leq 1$). Beyond this limit, the nonlinear dynamics dominated and the controller was unable to recover, resulting in divergence. Thus, while the linear feedback controller performs reliably for moderate deviations, there exists an upper bound on the initial error magnitude, beyond which convergence cannot be guaranteed.

This result demonstrate that the designed state-feedback controller effectively stabilizes the spacecraft near ω_{e1}, providing local asymptotic stability within a region consistent with the assumptions of linearization.

6 Control Design Scenario 2

In the second control design scenario, the equilibrium point $\omega_{e2} = [0, 1, 0]^T$ was chosen to demonstrate stabilization about a different principal axis. Controllability analysis showed that this equilibrium is also fully controllable (rank$(\mathcal{C}) = 3$), meaning that appropriate combinations of the control torques τ_1 and τ_2 can influence all three angular velocity components near this operating point.

The same linear feedback control structure as the Control Design Scenario 1,

$$u = -K(\omega - \omega_e)$$

was employed, with $u = [\tau_1, \tau_2]^T$ and $\omega_e = [0, 1, 0]^T$. Using the linearized model at this equilibrium and the same desired closed-loop poles -2.5, -3.0, and -3.5, MATLAB's place() command produced the gain matrix

$$K = \begin{bmatrix} 0.4767 & 0 & -1.7700 \\ 0 & 0.6679 & 0 \end{bmatrix}$$

The resulting closed-loop eigenvalues were -2.5, -3.0, and -3.5, confirming the intended dynamic response.

Simulations verified that the controller successfully stabilized the spacecraft around this new equilibrium. When initialized near ω_{e2}, the angular velocity components decayed exponentially to their target values, and the control torques quickly approached zero. The transient response was smooth and well-damped, consistent with the chosen pole placement. As the initial deviation increased, the system continued to converge until the applied torques reached their saturation limits, beyond which the linear feedback law was no longer valid.

Overall, the results confirm that both ω_{e1} and ω_{e2} can be made asymptotically stable through properly designed state-feedback control, while ω_{e3} remains uncontrollable. These findings highlight how feedback linearization and pole placement can effectively overcome natural instabilities in rigid-body rotational motion for multiple equilibrium orientations.

7 Future Directions

The results of this project demonstrated that linear state-feedback control based on pole placement can effectively stabilize the rotational dynamics of a rigid spacecraft around controllable equilibrium points. Both equilibria $\omega_{e1} = [1, 0, 0]^T$ and $\omega_{e2} = [0, 1, 0]^T$ were successfully stabilized, while $\omega_{e3} = [0, 0, 1]^T$ was shown to be uncontrollable due to the absence of torque authority along that axis. The simulations confirmed the theoretical predictions, illustrating asymptotic convergence and predictable limits imposed by actuator saturation.

For future work, the controller could be improved to handle larger deviations or stronger disturbances. One possible idea is to design a nonlinear controller that works beyond the region where the linear model is accurate. It could also be useful to test different pole locations or adjust the gains to reduce torque saturation. These steps would make the system more robust and realistic for broader conditions.